BLOCKCHAIN PROJECT

ProofPass

Blockchain-Based Internship Verification

A decentralized application for issuing, managing and independently verifying internship records on Ethereum Sepolia.

Technology: Solidity • Ethereum Sepolia • Remix IDE • MetaMask • Ethers.js • HTML/CSS/JavaScript

Project status: Smart contract deployed on Sepolia; frontend integrated and functionally tested.

1. Abstract

ProofPass is a blockchain-based internship verification decentralized application (DApp) designed to address the difficulty of independently verifying internship claims. The system enables authorized companies or organizations to issue verifiable internship records, associates those records with a student's wallet address, and allows students and recruiters to access verification information through a frontend connected to a deployed Solidity smart contract.

Each internship receives a unique identifier and contains core information such as the student wallet address, company name, role, start date, end date and project or work description. Authorized issuers can revoke a record when required, while users can verify a record using its internship ID. The frontend also provides status presentation, a shareable verification page and QR-code based access to the verification page.

The project was implemented with Solidity and deployed on the Ethereum Sepolia test network. Remix IDE was used for contract development and deployment, MetaMask for wallet-based transaction signing, and Ethers.js for frontend-to-contract integration.

The implementation satisfies the core requirements of the assigned ProofPass problem statement and provides a working end-to-end demonstration.

2. Problem Statement and Motivation

Students often include internship experience on resumes and professional profiles, but recruiters may have difficulty verifying whether an internship was actually completed. Traditional proofs such as certificates, screenshots or documents can communicate a claim without providing a simple, independently verifiable record.

The assigned ProofPass problem requires a blockchain-based internship verification system where authorized companies or organizations can issue verifiable internship records. Students must be able to view records associated with their wallet, recruiters or other users must be able to verify an internship using its ID, issuers must be able to revoke records, and the system must display whether a record is valid, revoked or completed.

The additional challenge is to generate a shareable verification page and QR code for every verified internship.

Challenges addressed

- Trusted issuer model so arbitrary wallets cannot create internship records.
- Unique internship identifiers for simple lookup and verification.
- Public verification that does not require the verifier to own the student's wallet.
- Revocation while keeping previous records traceable.
- A real frontend connected to the deployed smart contract rather than a contract-only prototype.

3. Objectives

Objective	Implementation / outcome
Issue verifiable records	Authorized issuers can create internship records through the DApp.
Unique identification	Each internship is represented by a unique internship ID.
Wallet association	The record stores the student's wallet address.
Public verification	Users can verify a record by entering its internship ID.
Student access	Students can view records associated with their connected wallet.
Revocation	Authorized issuers can invalidate records when required.
Status display	Frontend presents active/completed/revoked state.
Traceability	Records remain addressable for verification.
QR/shareable proof	Verification URL and QR access are provided.

4. Requirements Mapping

#	Required capability	ProofPass implementation
1	Authorized company/organization can issue	Issuer authorization + issueInternship()
2	Unique internship ID	Unique internship identifier
3	Required internship details	Student wallet, company, role, dates, work description
4	Only authorized issuers create	Access-control check
5	Students view records	getMyInternships() + wallet-connected frontend
6	Verification by ID	verifyInternship() / getInternship() + public UI
7	Issuer can revoke	revokeInternship()
8	Valid/revoked/completed display	Frontend status presentation
9	Previous records traceable	Records remain retrievable by ID
Extra	Shareable page + QR	Verification URL + QR code

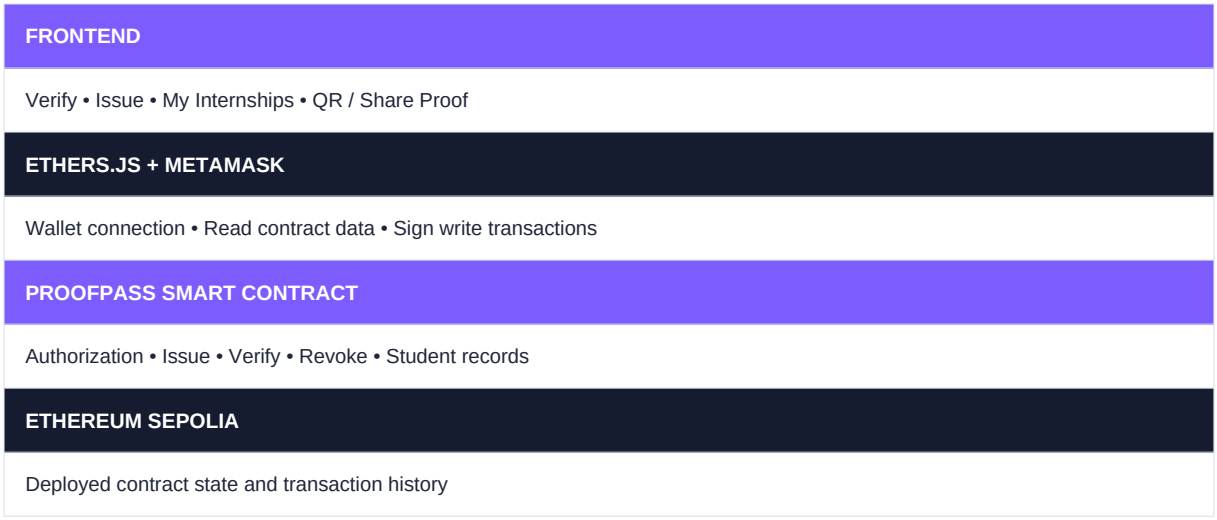
5. Proposed Solution

ProofPass uses a smart contract as the trusted record layer. Internship records are created through blockchain transactions and can be read through the deployed contract. The system separates responsibilities among issuers, students and verifiers.

Role	Responsibility
Issuer	Authorized organization/company wallet that creates or revokes internship records.
Student	Wallet associated with one or more internship records and able to view its records.
Verifier / Recruiter	Any user who needs to independently check an internship using its ID.
Smart Contract	Enforces authorization and validation rules and exposes internship data.
Frontend	Provides a human-friendly interface for issuing, viewing and verifying records.

6. System Architecture

The architecture consists of four logical layers: user interface, wallet/provider, blockchain contract and on-chain records.



Read operations can support public verification, while write operations require a connected wallet and an approved MetaMask transaction.

7. Smart Contract Design

The Solidity contract is the trust layer of ProofPass. It uses state variables, mappings/structs, access control and transaction validation.

Core internship record

Field	Purpose
Internship ID	Unique identifier used to retrieve and verify the record.
Student wallet	Blockchain address associated with the internship.
Company name	Organization that issued the internship.
Role	Internship position/title.
Start / end date	Internship period.
Work description	Project or work summary.
Issuer	Authorized wallet that created the record.
Issue timestamp	Time at which the record was issued.
Revocation state	Indicates whether the record has been invalidated.

Expected / implemented contract interface

Function	Purpose
issueInternship()	Create a new internship record.
verifyInternship()	Verify/retrieve an internship by ID.
revokeInternship()	Invalidate a record through an authorized issuer.
getInternship()	Read details of a specific internship.
getMyInternships()	Retrieve records associated with the connected student wallet.
authorizeIssuer()	Control which addresses can act as issuers.

8. Access Control, Validation and Security

Control	Purpose
Issuer authorization	Prevents arbitrary wallets from creating records.
Issuer-only revocation	Prevents unrelated users from invalidating a record.
Transaction validation	Rejects invalid state-changing requests.
Date validation	Prevents logically invalid internship periods.
Invalid ID handling	Nonexistent records are rejected/handled safely.
Wallet-based student access	Records are associated with a blockchain address.
Traceability	Existing records remain addressable, including revoked records.

ProofPass is deployed on the Sepolia test network for demonstration and development. A production deployment would require additional security auditing, privacy analysis, key-management controls and operational governance.

9. Technology Stack

Technology	Role
Solidity	Smart contract implementation.
Ethereum Sepolia	Test blockchain network for deployment and demonstration.
Remix IDE	Contract writing, compilation, deployment and direct testing.
MetaMask	Wallet connection and transaction signing.
Ethers.js	Frontend-to-smart-contract integration.
HTML	Frontend structure.
CSS	Visual styling and responsive interface.
JavaScript	Frontend logic and blockchain interaction.
QR generation	Creates QR access to the verification page.

10. Frontend and User Experience

The frontend was designed as a product-style verification interface rather than a minimal contract-testing page.

Interface	Experience
Verify Internship	Enter an ID and retrieve the blockchain-backed record.
Record card	Shows company, role, wallets, dates, issuer, issue date and work description.
Status badge	Communicates active/completed/revoked state.
Connect Wallet	Connects MetaMask for authenticated operations.
My Internships	Displays records associated with the connected student wallet.
Issue Internship	Lets an authorized issuer submit a new internship transaction.
QR code	Provides a scannable route to public verification.
Share proof	Creates a shareable verification link.

The verification screen was used to display a revoked record, demonstrating that the frontend reads and presents contract state rather than showing a static mock record.

11. Functional Workflows

A. Issuing an internship

- Issuer connects MetaMask.
- Frontend uses issuer authorization state.
- Issuer enters student wallet and internship details.
- Ethers.js sends the transaction through MetaMask.
- Smart contract validates and creates the record.
- The new internship ID becomes the verification reference.

B. Student viewing records

- Student connects the wallet associated with the records.
- Frontend requests the student's internship records.
- DApp displays proofs and their current status.

C. Recruiter verification

- Recruiter enters an internship ID or opens a shareable URL.
- Frontend reads the record from the deployed contract.
- Verification page displays internship details and validity.
- The same page can be reached through the QR code.

D. Revocation

- Authorized issuer signs the revocation transaction.
- Smart contract changes the revocation state.
- Future verification displays the record as revoked/invalid while the record remains traceable.

12. Testing and Validation

Test	Expected behavior	Observed result
Sepolia deployment	Contract deploys successfully	Successful deployment and frontend connection
Issue internship	Authorized issuer creates record	Tested successfully
Verify internship	Valid ID returns details	Tested successfully
Unauthorized issue	Non-authorized wallet rejected	Validation tested
Revocation	Authorized issuer invalidates record	Tested successfully
Revoked verification	Record shown revoked/invalid	Confirmed in frontend
Student records	Connected student sees records	Tested successfully
Invalid ID	Nonexistent record handled safely	Validation tested
Invalid dates	Invalid period rejected	Validation tested
QR / share proof	Verification route opens record	Implemented and tested

Testing focused on the minimum technical requirements: structured Solidity contract, access control, validation, Sepolia deployment, frontend connection, MetaMask integration, main functionality and transaction handling.

13. Deployment and Demonstration

Item	Project state
Network	Ethereum Sepolia testnet
Contract	ProofPass smart contract deployed and tested
Wallet	MetaMask used for connected operations
Frontend	Connected to deployed contract through Ethers.js
Production hosting	Ready for final static-hosting deployment

Recommended final demonstration

- Open the deployed frontend and verify an internship by ID.
- Connect the student wallet and open My Internships.
- Switch to an authorized issuer and issue a new internship.
- Show MetaMask transaction confirmation.
- Open the new verification page.
- Display and scan the QR code.
- Revoke an internship and demonstrate the changed status.

14. Results and Discussion

ProofPass demonstrates the complete lifecycle of a blockchain-backed internship record: creation by an authorized issuer, association with a student wallet, retrieval by a unique ID, independent verification, revocation and presentation of the resulting status through a frontend.

The project also addresses the additional challenge by connecting verification records to a shareable page and QR code. This creates a practical bridge between blockchain data and a recruiter-friendly verification experience.

A major design strength is that the verifier does not need to rely on a screenshot or private database. The verification interface reads the deployed smart contract and presents the record represented by its unique identifier.

Project outcome

- End-to-end DApp rather than a frontend-only prototype.
- Smart contract deployed on Ethereum Sepolia.
- Role-based issuance and revocation flow.
- Wallet-linked student records.
- Public verification by internship ID.
- Status-aware verification interface.
- QR and shareable verification experience.

15. Limitations and Future Scope

Current limitations

- Sepolia is a test network and is not a production deployment.
- Wallet addresses are public identifiers and are not a complete decentralized identity solution.
- The prototype does not yet implement enterprise issuer onboarding, legal credential standards or formal privacy controls.
- Production use would require independent smart-contract security auditing and stronger operational controls.

Future enhancements

- Decentralized identity and verifiable credential integration.
- Issuer organization profiles and multi-level authorization.
- Credential metadata standards and richer internship evidence.
- Employer/recruiter dashboards and verification analytics.
- Privacy-preserving references to supporting documents.
- Production deployment with stronger key management and governance.

16. Conclusion

ProofPass demonstrates how blockchain can be applied to a practical trust problem: verifying internship experience. The system uses a Solidity smart contract on Ethereum Sepolia as the verification layer and combines it with MetaMask, Ethers.js and a purpose-built frontend.

By restricting issuance to authorized issuers, assigning unique internship IDs, linking records to student wallets, supporting revocation and enabling public verification, ProofPass turns an internship claim into a record that can be independently checked through the DApp.

The project satisfies the central requirements of the assigned ProofPass problem statement and implements the additional QR-code and shareable verification challenge.

17. Appendix — Deliverables Checklist

Deliverable	Status
Solidity smart contract	Completed
Sepolia deployment	Completed
MetaMask integration	Completed
Ethers.js frontend integration	Completed
Issue internship	Completed / tested
Verify internship	Completed / tested
My Internships	Completed / tested
Revocation	Completed / tested
Status display	Completed / tested
Shareable verification page	Implemented
QR code	Implemented
Frontend UI	Implemented
Project presentation	Prepared
Final production hosting	Next deployment step

Primary requirements reference

Blockchain Project – Custom Problem Statements, Problem Statement 17: ProofPass — Blockchain-Based Internship Verification. The source defines the problem description, nine functional requirements, expected smart-contract functions, expected output and the additional shareable-page/QR challenge.

End of Report